

- **Who has issued the certificate?**

with

```
openssl s_client -connect polimi.it:443  
-servername polimi.it 2>/dev/null | openssl x509  
-noout -issuer
```

I find

```
issuer=C = GB, ST = Greater Manchester, L =  
Salford, O = Sectigo Limited, CN = Sectigo ECC  
Organization Validation Secure Server CA
```

Which means:

the CA is based in GB (Great Britain)

in the state of Great Manchester

and the locality is Salford;

the organization is **Sectigo Limited**

and the common name - that specifies the intermediate certificate
used to sign the certificate for polimi.it - is

Sectigo ECC Organization Validation Secure Server CA

- **For what exact domain it has been certificated?**

with

```
openssl s_client -connect example.com:443  
-servername example.com 2>/dev/null | openssl  
x509 -noout -subject -ext subjectAltName
```

I find

```
subject=C = IT, ST = Milano, O = Politecnico di  
Milano, CN = 2023.polimi.it  
X509v3 Subject Alternative Name:  
    DNS:2023.polimi.it,  
DNS:dynamicpoli.polimi.it, DNS:polimi.it,  
DNS:www-polimi.it, DNS:www.english.polimi.it,  
DNS:www.polimi.it, DNS:www1.polimi.it,  
DNS:wwwpolimi.it
```

So, the certificate country is Italy

The Politecnico di Milano organization owns the certificate, and is based in Milan

- **When does it expire?**

with

```
openssl s_client -connect polimi.it:443  
-servername polimi.it 2>/dev/null | openssl x509  
-noout -enddate
```

I find

```
notAfter=Dec  3 23:59:59 2025 GMT
```

which is clearly the expiration date

- **What is the encryption algorithm and key size?**

With

```
openssl s_client -connect polimi.it:443 -servername polimi.it 2>/dev/  
null | openssl x509 -noout -text | grep -E "Signature Algorithm|  
Public-Key"
```

I find

```
Signature Algorithm: ecdsa-with-SHA256  
Public-Key: (256 bit)  
Signature Algorithm: ecdsa-with-SHA256
```

which means the algorithm used is ECDSA (Elliptic Curve Digital Signature Algorithm) with hash function SHA-256;
the key is 256 bit long.

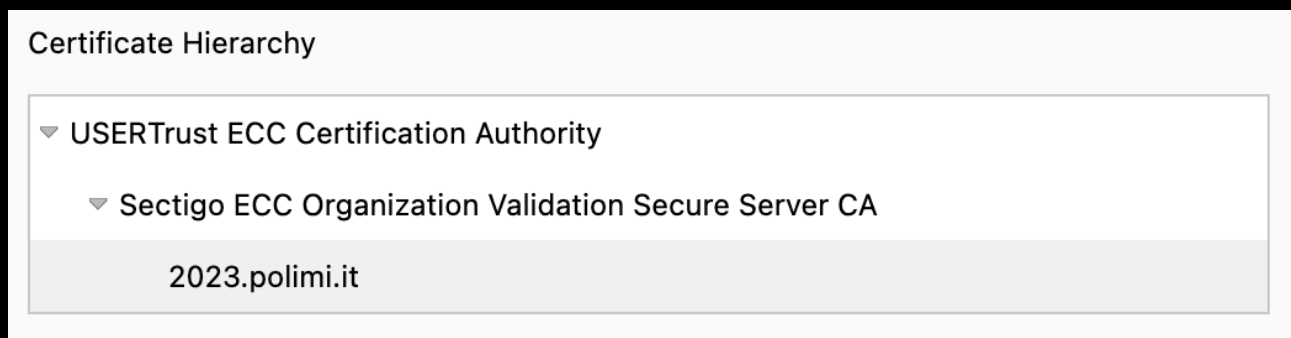
- **Manual validation with openssl verify with local file:**

With

```
openssl s_client -connect polimi.it:443  
-servername polimi.it </dev/null 2>/dev/null |  
openssl x509 -outform PEM > polimi.crt
```

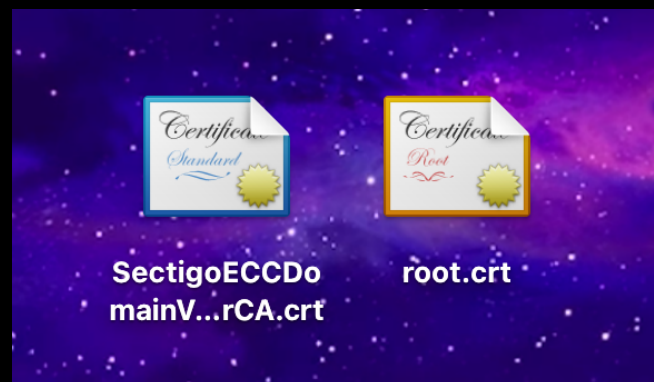
a file named polimi.crt is downloaded.

Looking at the Certificate Hierarchy of the site, this is shown:



It is possible to download the other two certificates directly from the websites:

and to concatenate them into a new .crt file through



```
cat SectigoECCDomainValidationSecureServerCA.crt  
root.crt > combined.crt
```

At this point, the hierarchy can be verified with

```
openssl verify -CAfile combined.crt polimi.crt
```

Finding some errors, I tried with

```
cat polimi.crt  
SectigoECCDomainValidationSecureServerCA.crt  
root.crt > combined.crt
```

but I still got

```
error 20 at 0 depth lookup: unable to get local  
issuer certificate  
error polimi.crt: verification failed
```

I tried to change folders, repeat the process, but I couldn't have a positive answer returned.

All the files format has been checked with 'cat' command and the format was correct (-----BEGIN CERTIFICATE----- etc...)